Lecture 12

Thursday March 5th 2026

Start reading Pintos documentation for project 2

Recap

- synchronization is necessary for any type of parallel programming
- Memory is shared so problems are introduced when reading/writing
  - two threads accessing a variable at the same time is a problem
  - need to implement serial access
- Producer/consumer model
- Race conditions are a problem

Synchronization Hardware

- many systems provide hardware support for critical section code

Semaphores

- TSL mean that you have to constantly check for a lock which will use a lot of the CPU time
- Semaphores are a synchronization tool for CS problems
- Semaphore S - int variable
- Can only be accessed through two standard operations:
  - `wait()` or `P(`
    - if free -> enter
    - if blocked -> stop
  - `sign)al()` or `V()`
    - sends a signal that the process is done with the resource
- Classical implementation uses busy-waiting (while loop to block programs - not efficient)
- Review recording of the lecture for example

Problems of Synchronization

- Bounded buffer problem
- Readers and writers problem
- Dining-Philosophers problem